Lecture Notes

Lecture 15

Perturbation Theory for Linear Operators

This optional capstone asks what remains when an operator cannot be diagonalized exactly but is close to one that can. Starting from the spectral decomposition of Lecture 10 and the norm, spectrum, and resolvent language of Lecture 14, we study self-adjoint families

$$H(\varepsilon) = H_0 + \varepsilon V.$$

The spectral gap controls both what can be proved and what can be approximated. No core lecture depends on this material, and it is outside the mandatory assessment unless an instructor explicitly includes it.

Learning objectives

By the end of this lecture you will be able to:

- Use the operator norm and resolvent to bound how far a spectrum can move.
- Compute first- and second-order shifts of a simple eigenvalue and the first-order change of its eigenvector.
- State and apply the Hellmann–Feynman identity.
- Diagnose the failure of non-degenerate formulas and split a degenerate level by diagonalizing PVP.
- Explain which perturbative ideas survive, and which can fail, in infinite dimensions.

1 Spectral size and the resolvent

Lecture 14 defined the operator norm and resolvent for bounded operators. In the finite-dimensional normal case, Lecture 10's spectral projections make both quantities completely explicit.

One more quantity is read straight off the spectrum. How much can an operator stretch a vector?

Definition 15.1 (Operator norm). For $T \in \mathcal{L}(V)$ on an inner product space, the *operator norm* is $\|T\| := \sup_{v\neq 0}\|Tv\|/\|v\| = \max_{\|v\|=1}\|Tv\|$, finite in finite dimensions (a continuous function maximized on the unit sphere). At once $\|Tv\| \le \|T\|\,\|v\|$, and hence $\|ST\| \le \|S\|\,\|T\|$.

Proposition 15.2 (Norm and resolvent of a normal operator). Let $T = \sum_k \lambda_k P_k$ be normal. Then

$$\|T\| = \max_k |\lambda_k|,$$

and for every $z \notin \operatorname{spec}(T)$ the operator $T - zI$ is invertible with

$$\|(T - zI)^{-1}\| = \frac{1}{\operatorname{dist}\big(z, \operatorname{spec}(T)\big)}, \qquad \operatorname{dist}\big(z, \operatorname{spec}(T)\big) = \min_k |\lambda_k - z|.$$

Proof. Resolve $v = \sum_k P_k v$ into orthogonal eigen-components. By Pythagoras (Lecture 7),

$\|v\|^2 = \sum_k \|P_k v\|^2$ and

$$\|Tv\|^2 = \sum_k |\lambda_k|^2 \|P_k v\|^2 \le \left(\max_k |\lambda_k|\right)^2 \|v\|^2,$$

with equality when v is an eigenvector for a largest-modulus eigenvalue; hence $\|T\| = \max_k |\lambda_k|$. For the second claim, the functional calculus applied to $f(\lambda) = (\lambda - z)^{-1}$ gives the normal operator $(T - zI)^{-1} = \sum_k (\lambda_k - z)^{-1} P_k$, whose norm is therefore $\max_k |\lambda_k - z|^{-1} = \left(\min_k |\lambda_k - z|\right)^{-1}$. ∎

The resolvent $(T - zI)^{-1}$ thus blows up exactly as z approaches the spectrum, at a rate set by nothing but the distance to the nearest eigenvalue.

2 Perturbing a spectral decomposition

Exact diagonalization is the exception, not the rule. Almost every operator a physicist actually meets has the shape $H(\varepsilon) = H_0 + \varepsilon V$, with H_0 self-adjoint and already diagonalized, V self-adjoint, and $\varepsilon \in \mathbb{R}$ small: a spin in a slightly tilted field, an atom in a weak electric field, a lattice with one defective site. Two questions, in this order. How far can the spectrum move at all? And, knowing it moves only a little, what is the shift to leading orders in ε? Throughout, a denominator is written with the level of interest first, $E_n - E_m$.

How far can the spectrum move?

The first question has a clean answer, with Proposition 15.2 as the engine. It needs one standard tool, the operator version of $\frac{1}{1-x} = 1 + x + x^2 + \cdots$.

Lemma 15.3 (Neumann series). If $X \in \mathcal{L}(V)$ satisfies $\|X\| < 1$, then $I - X$ is invertible and

$$(I - X)^{-1} = \sum_{n \ge 0} X^n, \qquad \|(I - X)^{-1}\| \le \frac{1}{1 - \|X\|}.$$

Proof. Sub-multiplicativity (Definition 15.1) gives $\|X^n\| \le \|X\|^n$, so the series converges absolutely in the finite-dimensional space $\mathcal{L}(V)$. Its partial sums satisfy $(I - X)S_N = S_N(I - X) = I - X^{N+1} \to I$, since $\|X^{N+1}\| \le \|X\|^{N+1} \to 0$; hence the limit S obeys $(I - X)S = S(I - X) = I$, with $\|S\| \le \sum_n \|X\|^n = (1 - \|X\|)^{-1}$. ∎

Theorem 15.4 (Spectral stability). Let H_0 and V be self-adjoint, $\varepsilon \in \mathbb{R}$, and let $E_1, \dots, E_r$ be the distinct eigenvalues of H_0. Then

$$\operatorname{spec}\big(H_0 + \varepsilon V\big) \subseteq \bigcup_{k=1}^{r} \big\{ z \in \mathbb{C} : |z - E_k| \le |\varepsilon| \, \|V\| \big\}.$$

Proof. Let z satisfy $\operatorname{dist}(z, \operatorname{spec} H_0) > |\varepsilon| \|V\|$; we show $H_0 + \varepsilon V - zI$ is invertible. First, $z \notin \operatorname{spec}(H_0)$, so $H_0 - zI$ is invertible, and we may factor

$$H_0 + \varepsilon V - zI = (H_0 - zI)\big(I + \varepsilon (H_0 - zI)^{-1} V\big).$$

Since H_0 is normal, Proposition 15.2 and sub-multiplicativity give

$$\left\| -\varepsilon (H_0 - zI)^{-1} V \right\| \le |\varepsilon| \, \|(H_0 - zI)^{-1}\| \, \|V\| = \frac{|\varepsilon| \, \|V\|}{\operatorname{dist}(z, \operatorname{spec} H_0)} < 1,$$

so the second factor is invertible by Lemma 15.3. A product of invertible operators is invertible, so $z \notin \operatorname{spec}(H_0 + \varepsilon V)$. ∎

Every eigenvalue therefore stays within $|\varepsilon|\|V\|$ of an unperturbed one: the spectrum is *Lipschitz* in the perturbation—no smallness assumption, no series. The work was done by the resolvent bound of Proposition 15.2, which is exactly where self-adjointness entered.

> **Proposition 15.5 (No level is lost · cited).** If $2|\varepsilon|\|V\| < \min_{j\neq k}|E_j - E_k|$, the disks of Theorem 15.4 are pairwise disjoint, and each contains exactly as many eigenvalues of $H_0 + \varepsilon V$, with multiplicity, as H_0 has at its centre.

A small perturbation thus moves levels without destroying them or letting two groups merge, provided the disks stay apart. The controlling ratio is *perturbation size over spectral gap*—a theme that returns in every formula below; Example 15.10 works it out.

First- and second-order corrections

Now assume every eigenvalue of H_0 is *simple* (non-degenerate), with $H_0|m\rangle = E_m|m\rangle$ for an orthonormal basis $\{|m\rangle\}$ and eigenprojections $P_m = |m\rangle\langle m|$. Fix a level $|n\rangle$.

> **Theorem 15.6 (Analyticity of a simple level · sketched).** If E_n is a simple eigenvalue of H_0, then for small $|\varepsilon|$ the operator $H(\varepsilon)$ has exactly one eigenvalue $E_n(\varepsilon)$ near E_n, and it—with a suitable choice of eigenvector—is given by a convergent power series in ε.

Sketch. Uniqueness is Proposition 15.5. For the series, apply the implicit function theorem to $p(\lambda,\varepsilon) = \det\big(\lambda I - H(\varepsilon)\big)$, a polynomial in both arguments: $p(E_n, 0) = 0$ and $\partial p/\partial\lambda \neq 0$ there, *precisely because* E_n is a simple root, giving a unique analytic branch through E_n. That settles the eigenvalue but says nothing about the eigenvector, so apply the same theorem a second time, to the pair

$$F(v,\lambda,\varepsilon) = \big(H(\varepsilon)v - \lambda v,\ \langle n\,|\,v\rangle - 1\big),$$

which vanishes at $(|n\rangle, E_n, 0)$. Its derivative there in (v,λ) sends (w,μ) to $\big((H_0 - E_n)w - \mu|n\rangle,\ \langle n\,|\,w\rangle\big)$; pairing the first component with $\langle n|$ kills the left-hand side and forces $\mu = 0$, after which simplicity of E_n gives $w \in \operatorname{span}(|n\rangle)$ and $\langle n\,|\,w\rangle = 0$ gives $w = 0$. The derivative is therefore injective, hence invertible, and $|n(\varepsilon)\rangle$ is analytic as well—normalized, as it happens, exactly as (15.1) requires. ■

This licenses the ansatz; without it what follows would be formal. Write

$$E_n(\varepsilon) = E_n + \varepsilon E_n^{(1)} + \varepsilon^2 E_n^{(2)} + \cdots, \qquad |n(\varepsilon)\rangle = |n\rangle + \varepsilon|n^{(1)}\rangle + \varepsilon^2|n^{(2)}\rangle + \cdots, \tag{15.1}$$

normalizing by $\langle n\,|\,n(\varepsilon)\rangle = 1$, i.e. $\langle n\,|\,n^{(k)}\rangle = 0$ for $k \geq 1$ (the eigenvector is fixed only up to scale; this is the convenient choice). Substituting into $H(\varepsilon)|n(\varepsilon)\rangle = E_n(\varepsilon)|n(\varepsilon)\rangle$, the terms of order ε and ε^2 read

$$(H_0 - E_n)|n^{(1)}\rangle = \big(E_n^{(1)} - V\big)|n\rangle, \qquad (H_0 - E_n)|n^{(2)}\rangle = \big(E_n^{(1)} - V\big)|n^{(1)}\rangle + E_n^{(2)}|n\rangle. \tag{15.2}$$

Pair the first with $\langle n|$: the left side vanishes, since H_0 is self-adjoint with E_n real, so $\langle n|\,(H_0 - E_n)\,|n^{(1)}\rangle = \langle (H_0 - E_n)n, n^{(1)}\rangle = 0$, while the right side is $E_n^{(1)} - \langle n|\,V\,|n\rangle$. Pairing it with $\langle m|$ for $m \neq n$ gives $(E_m - E_n)\langle m\,|\,n^{(1)}\rangle = -\langle m|\,V\,|n\rangle$. Pairing the second with $\langle n|$ kills its left side for the same reason, leaving $0 = -\langle n|\,V\,|n^{(1)}\rangle + E_n^{(2)}$. Together:

$$\boxed{E_n^{(1)} = \langle n|\,V\,|n\rangle, \qquad |n^{(1)}\rangle = \sum_{m\neq n}\frac{\langle m|\,V\,|n\rangle}{E_n - E_m}|m\rangle, \qquad E_n^{(2)} = \sum_{m\neq n}\frac{|\langle m|\,V\,|n\rangle|^2}{E_n - E_m}} \tag{15.3}$$

(the last using $\langle n|V|m\rangle = \overline{\langle m|V|n\rangle}$, since V is self-adjoint). These are the *Rayleigh–Schrödinger* formulas: a level shifts by the diagonal matrix element of V in its own state, the state picks up an admixture of every other state weighted by coupling over gap, and at second order the level shifts by a sum over all other levels, coupling squared over gap.

Remark 15.7 (The reduced resolvent). Formula (15.3) looks basis-dependent but is not. With the *reduced resolvent* $S_n := \sum_{m\neq n} P_m/(E_n - E_m)$—the inverse of $E_n - H_0$ on the orthogonal complement of $|n\rangle$—it reads

$$|n^{(1)}\rangle = S_n V|n\rangle, \qquad E_n^{(2)} = \langle n|VS_nV|n\rangle,$$

written entirely in the eigenprojections from the spectral-decomposition theorem of Lecture 10. This is the form that extends beyond finite dimensions, where S_n is built from the resolvent $(H_0 - z)^{-1}$ rather than from a finite sum.

Remark 15.8 (Level repulsion, and two free checks). Every denominator in $E_n^{(2)}$ is negative for the *lowest* level and positive for the highest, so second order always pushes the ground state down and the top state up: levels repel. Two identities check any calculation—summing the first of (15.3) gives $\sum_n E_n^{(1)} = \operatorname{tr} V$, and pairing the (m,n) and (n,m) terms of the third, whose denominators differ only in sign, gives $\sum_n E_n^{(2)} = 0$. Both are forced by $\operatorname{tr} H(\varepsilon) = \operatorname{tr} H_0 + \varepsilon \operatorname{tr} V$ being exactly linear in ε.

Remark 15.9 (When is the perturbation small?). The expansion parameter is not ε but the dimensionless ratio $\varepsilon|\langle m|V|n\rangle|/|E_n - E_m|$: coupling measured against the *gap*. A nominally tiny ε buys nothing when two levels are nearly degenerate—the situation the next subsection addresses. (The companion identity $dE_n/d\varepsilon = \langle n(\varepsilon)|V|n(\varepsilon)\rangle/\langle n(\varepsilon)\,|\,n(\varepsilon)\rangle$, the *Hellmann–Feynman* theorem, is set as an exercise. The denominator is 1 for a unit eigenvector, but not for the intermediate-normalized $|n(\varepsilon)\rangle$ of (15.1), whose norm is $1 + O(\varepsilon^2)$.)

Example 15.10 (A three-level system, worked out). Take, in the standard basis $|1\rangle, |2\rangle, |3\rangle$,

$$H_0 = \operatorname{diag}(0,1,3), \qquad V = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & -3 \end{pmatrix}.$$

First, stability. The eigenvalues of V are $-3.3460,\ 0.4067,\ 2.9392$, so $\|V\| = 3.3460$ by Proposition 15.2, while the smallest gap of H_0 is 1. Proposition 15.5 therefore applies while $2|\varepsilon|(3.3460) < 1$, i.e. $|\varepsilon| < 0.149$: up to there the three levels sit in three disjoint disks and can be tracked one by one.

Now the shifts. To first order they are the diagonal entries of V,

$$E_1^{(1)} = 2, \qquad E_2^{(1)} = 1, \qquad E_3^{(1)} = -3,$$

summing to $0 = \operatorname{tr} V$ as they must. All off-diagonal entries equal 1, so the second-order shifts are

$$E_1^{(2)} = \tfrac{1}{0-1} + \tfrac{1}{0-3} = -\tfrac{4}{3}, \qquad E_2^{(2)} = \tfrac{1}{1-0} + \tfrac{1}{1-3} = \tfrac{1}{2}, \qquad E_3^{(2)} = \tfrac{1}{3-0} + \tfrac{1}{3-1} = \tfrac{5}{6},$$

summing to $-\frac{4}{3} + \frac{1}{2} + \frac{5}{6} = 0$, the second check. Read physically, the lowest level is pushed *down* by both partners, the highest *up* by both, and the middle one, with a partner on each side, barely moves. For the ground state $E_1(\varepsilon) \approx 2\varepsilon - \frac{4}{3}\varepsilon^2$, against the exact eigenvalue:

ε	first order	second order	exact
0.05	0.1	0.096667	0.096562
0.10	0.2	0.186667	0.185892
0.20	0.4	0.346667	0.341699

Doubling ε multiplies the first-order error by about 4 and the second-order error by about 8—the errors are $O(\varepsilon^2)$ and $O(\varepsilon^3)$, as (15.1) predicts—and the accuracy visibly deteriorates at $\varepsilon = 0.2$, just past the 0.149 threshold beyond which the disjoint-disk guarantee no longer applies. The first-order *state* is equally concrete: (15.3) gives $|1(\varepsilon)\rangle \approx (1, -0.1, -0.0333)$ at $\varepsilon = 0.1$. Scaled to the same intermediate normalization $\langle 1 \,|\, 1(\varepsilon)\rangle = 1$, the exact eigenvector is $(1, -0.1055, -0.0356)$—the leading component agrees identically, by the normalization, and the two admixtures are right to within about 6%. (Its unit-normalized form is $(0.9939, -0.1049, -0.0354)$; comparing against *that* would charge the approximation for a change of normalization.) ◁

Degenerate levels

Formula (15.3) divides by zero the moment two levels coincide, and Theorem 15.6 does not apply, having needed a *simple* root. This is no technical blemish: a degenerate level does not shift, it *splits*. The resolution is a reading of the degenerate-observable example from Lecture 10, where a degenerate eigenvalue was seen to leave freedom in the eigenbasis—freedom fixed in physics by a further commuting observable. A perturbation fixes it too, and in the only way it can.

> **Theorem 15.11 (First-order splitting · sketched).** Let E_0 be an eigenvalue of H_0 of multiplicity d, with eigenprojection P. To first order the level splits into $E_0 + \varepsilon\mu_1, \ldots, E_0 + \varepsilon\mu_d$, where the μ_j are the eigenvalues of the *compressed* operator PVP restricted to range P—a $d \times d$ self-adjoint matrix—and the eigenvectors of PVP are the correct zeroth-order states.

Sketch. Granting that the perturbed eigenvalue and eigenvector admit expansions (15.1) about some $|\psi\rangle \in$ range P (for a self-adjoint family this is Rellich's theorem, quoted here), the order-ε equation reads $(H_0 - E_0)|\psi^{(1)}\rangle = \big(E^{(1)} - V\big)|\psi\rangle$. Apply P: since H_0 commutes with P and $(H_0 - E_0)P = 0$, the left side vanishes, leaving $0 = E^{(1)}|\psi\rangle - PVP|\psi\rangle$ (using $P|\psi\rangle = |\psi\rangle$). So $|\psi\rangle$ is an eigenvector of PVP on range P with eigenvalue $E^{(1)}$; and PVP being self-adjoint on the d-dimensional range P, the spectral theorem supplies d orthonormal such states with real eigenvalues. ■

In one sentence: *a degeneracy is lifted by the part of the perturbation acting inside the degenerate eigenspace*, and the correct zeroth-order states are those diagonalizing it—which is why a physicist picks "good quantum numbers" to diagonalize the perturbation before perturbing. Zeeman and Stark splitting are precisely this calculation.

Example 15.12 (A degeneracy split). Take $H_0 = \operatorname{diag}(1, 1, 4)$ and V with rows $(0, 2, 1)$, $(2, 0, 1)$, $(1, 1, 0)$. The non-degenerate formula is meaningless here: $\langle 1|\,V\,|2\rangle = 2 \neq 0$ while $E_1 - E_2 = 0$. Instead compress V onto range $P = \operatorname{span}(|1\rangle, |2\rangle)$:

$$PVP\big|_{\text{range}\,P} = \begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix} = 2\sigma_x, \qquad \mu = \pm 2,$$

so the degenerate level splits as $E \approx 1 \pm 2\varepsilon$, with correct zeroth-order states $|\pm\rangle = \frac{1}{\sqrt{2}}(|1\rangle \pm |2\rangle)$—the σ_x eigenstates from the Pauli-σ_x example in Lecture 10. At $\varepsilon = 0.1$ the exact eigenvalues are $0.800000,\ 1.192875,\ 4.007125$ against the predicted $0.8,\ 1.2,\ 4$.

Two things are worth noticing. The lower branch $1 - 2\varepsilon$ is *exact to all orders*, for a reason worth more than any series: $\langle 3|\,V\,|-\rangle = 0$, so $|-\rangle$ is an exact eigenvector of $H(\varepsilon)$ for every ε. And once the degeneracy is lifted the levels are simple again, so ordinary second-order theory resumes—for the upper branch only $|3\rangle$ is left to sum over, giving $E_+^{(2)} = |\langle 3|\,V\,|+\rangle|^2/(1-4) = -\frac{2}{3}$ and $1 + 2\varepsilon - \frac{2}{3}\varepsilon^2 = 1.193333$ at $\varepsilon = 0.1$, against the exact 1.192875. ◁

3 Beyond finite dimensions

The same factorization also gives a resolvent expansion. With $R_0(z) = (H_0 - z\,\mathrm{id})^{-1}$,

$$\left(H_0 + \varepsilon V - z\,\mathrm{id}\right)^{-1} = \sum_{n\geq 0} (-\varepsilon)^n R_0(z)\left(V R_0(z)\right)^n,$$

whenever $|\varepsilon|\,\|V R_0(z)\| < 1$. In finite dimensions its poles track eigenvalues. In Hilbert space the spectrum may also be continuous and the operators may be unbounded, so additional hypotheses become load-bearing.

Remark 15.13 (Perturbation theory in infinite dimensions). The finite-dimensional stability argument transfers widely; the formal series is what breaks. Spectral stability survives unchanged for bounded self-adjoint operators, because its only ingredient—that $\|(T - z\,\mathrm{id})^{-1}\| = 1/\mathrm{dist}(z, \mathrm{spec}\, T)$ for normal T—is a consequence of the spectral theorem, not of finite dimensionality. The Rayleigh–Schrödinger formulas likewise remain valid for an isolated eigenvalue of finite multiplicity, even when V is unbounded, provided V is H_0-bounded with relative bound less than one—the *Kato–Rellich* theorem, which covers the Coulomb potential. Three things genuinely change.

- *The sum becomes an integral.* The reduced resolvent $S_n = \sum_{m\neq n} P_m/(E_n - E_m)$ runs over the whole spectrum, so where a continuous part is present it is an integral $\int (E_n - \lambda)^{-1} dP(\lambda)$ against the projection-valued measure of the spectral theorem of Lecture 14.
- *The bound may be unavailable.* Most physical perturbations have $\|V\| = \infty$, so the disk estimate proved above says nothing and relative boundedness must replace it.
- *An eigenvalue can dissolve.* An eigenvalue embedded in continuous spectrum may cease to be an eigenvalue for every $\varepsilon \neq 0$, becoming a *resonance* with a finite lifetime (Fermi's golden rule). The eigenvector does not merely move; it stops existing—with no finite-dimensional analogue.

Example 15.14 (A divergent series that works). The quartic anharmonic oscillator $H(\varepsilon) = \frac{1}{2}(\hat{p}^2 + \hat{x}^2) + \varepsilon\hat{x}^4$ has a Rayleigh–Schrödinger series with *zero* radius of convergence, yet its first few terms are numerically excellent. The reason is visible without any computation: for $\varepsilon < 0$ the quartic term drives the potential to $-\infty$, so $H(\varepsilon)$ is unbounded below and has no discrete spectrum at all. No disk about $\varepsilon = 0$ can be a domain of analyticity, however small. The series is *asymptotic* rather than convergent—the normal situation in quantum field theory, and the reason "perturbation theory works" and "the perturbation series converges" are different claims. ◁

Summary of main concepts

Perturbation theory is controlled by coupling relative to a spectral gap. Resolvent bounds first certify that the spectrum stays nearby. For an isolated simple level, Rayleigh–Schrödinger theory then gives the first two energy corrections and the first change in the eigenstate. When a level is degenerate, one must first diagonalize the compressed perturbation PVP. Infinite-dimensional problems retain the resolvent viewpoint but can lose boundedness, isolated eigenvalues, and convergence of the formal series.

- **Operator norm and resolvent**—for normal T, $\|T\| = \max_k |\lambda_k|$ and $\|(T - zI)^{-1}\| = 1/\mathrm{dist}(z, \mathrm{spec}\, T)$.
- **Spectral stability**—$\mathrm{spec}(H_0 + \varepsilon V)$ stays within $|\varepsilon|\|V\|$ of $\mathrm{spec}(H_0)$.
- **Non-degenerate perturbation**—$E_n^{(1)} = \langle n|\, V\, |n\rangle$ and $E_n^{(2)} = \sum_{m\neq n} |\langle m|\, V\, |n\rangle|^2/(E_n - E_m)$.

- **State mixing**—other eigenstates enter with amplitude equal to coupling divided by spectral gap.
- **Degenerate perturbation**—a degenerate level splits according to the eigenvalues of PVP on its eigenspace.
- **Infinite-dimensional caution**—unbounded perturbations, continuous spectrum, resonances, and asymptotic series require more than finite matrix algebra.